Volantex Ranger fixed wing INAV 8.0.1, SpeedyBee F405 Wing)

by Achraf

Volantex RC Ranger Hardware Overview

Pusher motor:
3715 powerful out runner brushless motor

VTX + Camera

GPS + Compass

ESC:
Easy-Plug 40A Switch mode BEC brushless ESC



Radio control:
ELRS 915 Radio

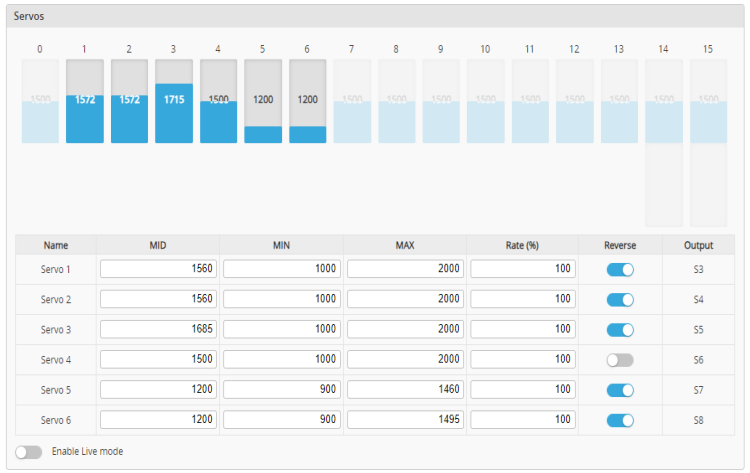
Power Source :
11.1V/14.8V 1800mAh ~ 10000mAh Li-Po

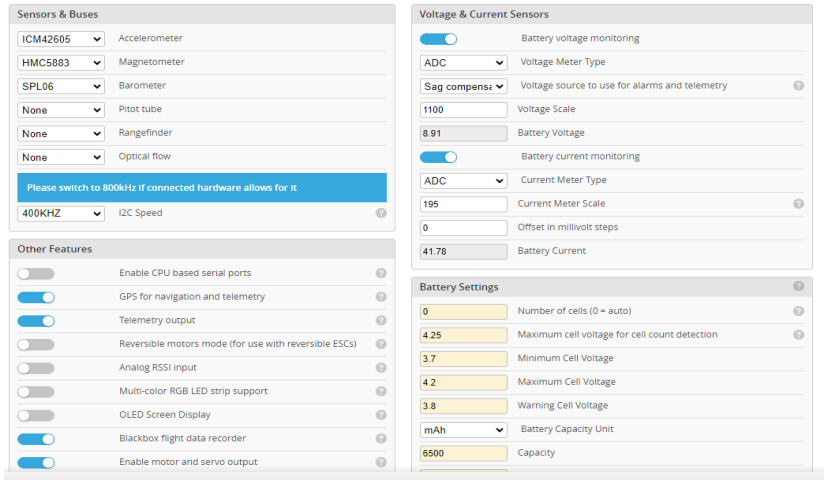
Flight Controller
Speedybee F405 INAV

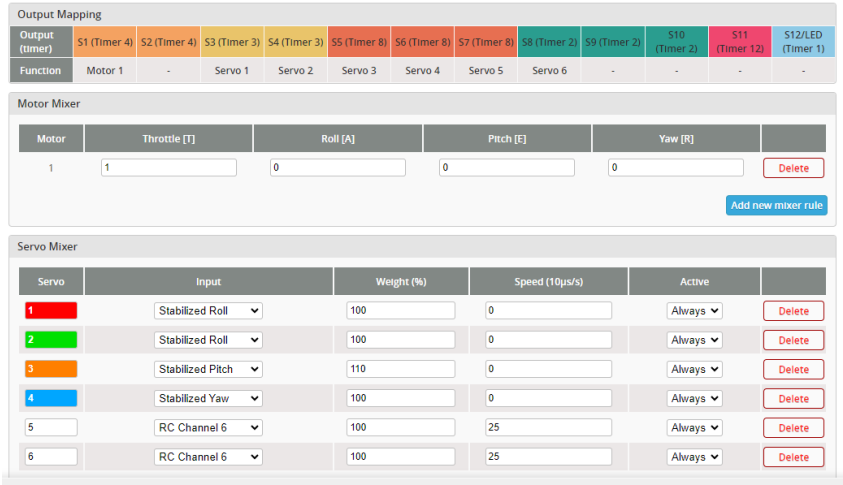
Servos: 9-gram Servo * 6pcs

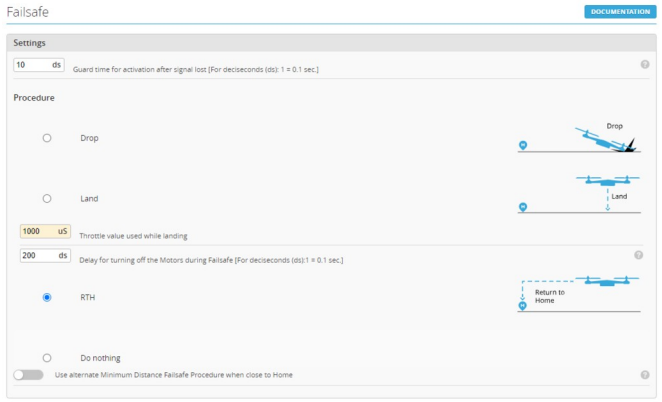
Propeller:
1060 Propeller

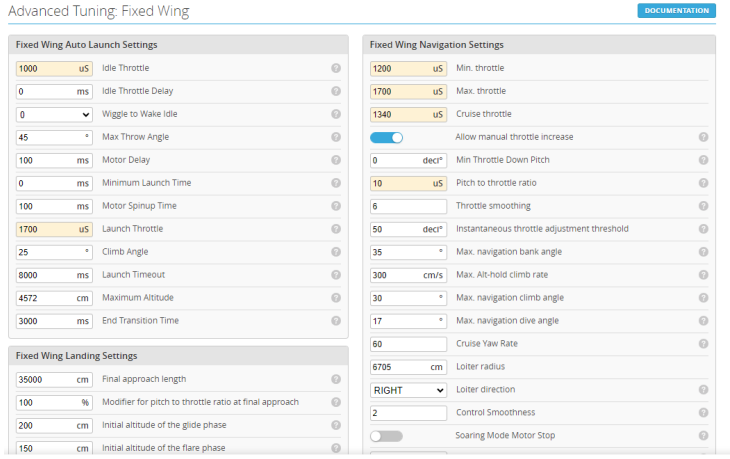
Volantex RC Ranger Software Overview

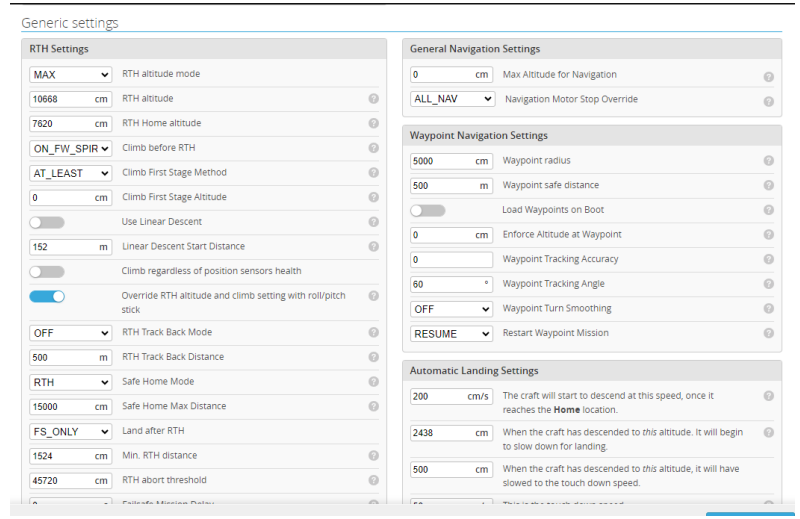












Ranger Aircraft: Yaw Drift in Fixed-Wing iNav (Pusher)

the Volantex Rangeer iNav-controlled push-prop (pusher) planes like r “yaw” continuously to the right in any stabilized mode (GPS/course hold, Angle, etc.) even though the aircraft holds level.





Yawing continuously to the right

Symptoms

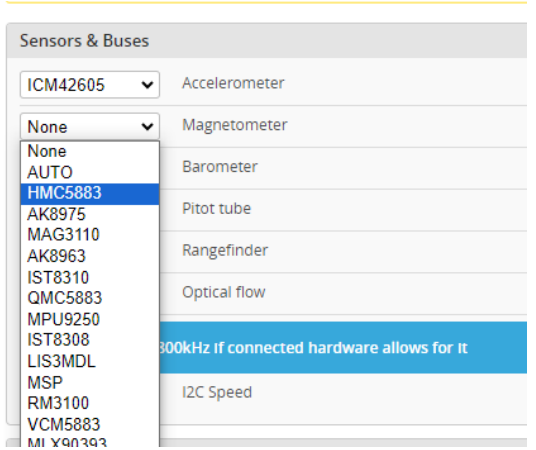
Holds level but slowly yaws right in ANGLE/HORIZON/NAV; will circle unless left rudder held.Course/Heading hold doesn’t stop the yaw.

What I see in your config (diff-all)

MAG enabled (HMC5883) and board alignment offsets set (roll=30°, pitch=10°).

Very strong yaw FF (fw_ff_yaw = 255) and high yaw P/D.

Yaw deadband set; small-angle = 180 (ok).



Most likely causes (in order)

1.Heading reference fighting you (MAG misalignment/interference).

2.Yaw stabilization too aggressive (large Yaw FF / PIDs).

3.Rudder sign/mix wrong or too much authority.

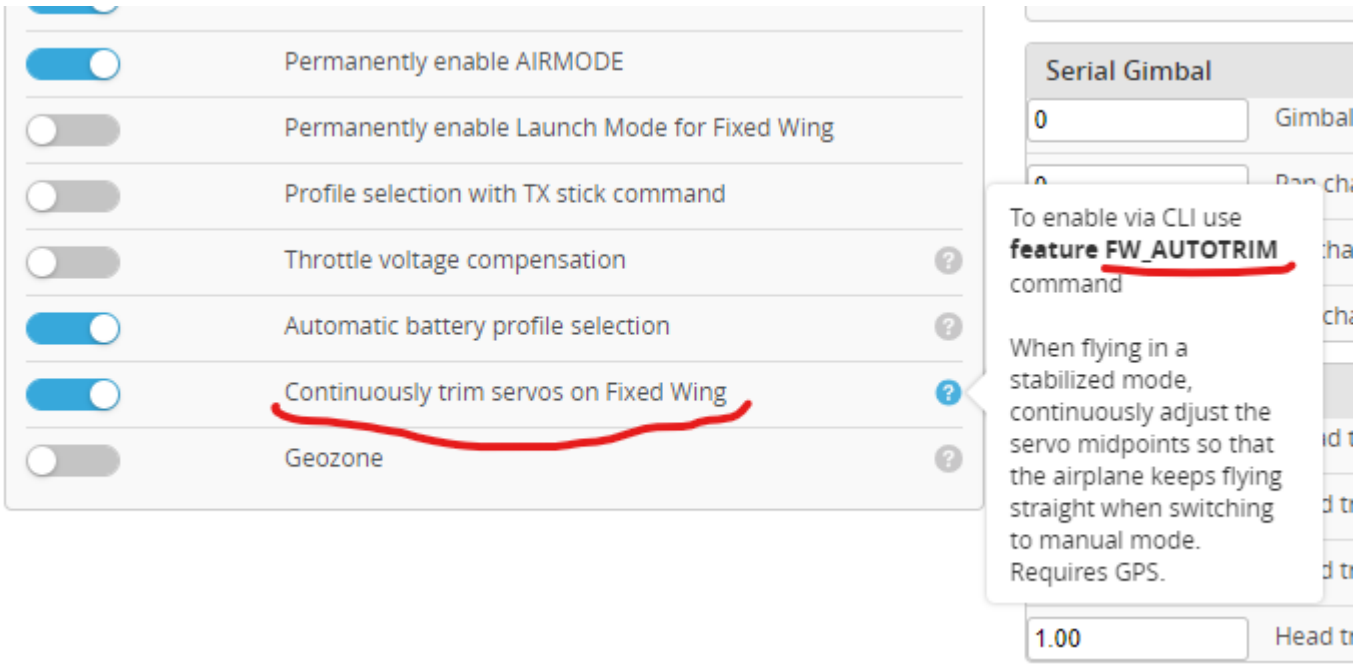
4.Airframe bias (pusher thrust line/fin not perfectly square).

5.Board alignment not matching the fuselage.



Compass (MAG) Issues

In fixed-wing iNav, the compass is normally only used when GPS ground speed is unavailable,



A misaligned or noisy magnetometer can induce a constant heading error.

Fix plan (do these in order)

1. Disable MAG for fixed-wing (use GPS course only):

Configuration>Sensors: disable Magnetometer (or set mag_hardware = NONE), Save & Reboot.

Fly ANGLE at mid-throttle and see if yaw drift disappears.

2. Make yaw manual (no auto-rudder) while we diagnose:

PID tab: set Yaw P = 0, I = 0, D = 0, FF = 0 (or in Mixer set RC Yaw / disable Stabilized Yaw).

Bench check: in ANGLE, twist nose right → rudder must move left. If not, reverse in Servos (don't reverse in TX).

3. Reduce GPS-mode rudder authority:

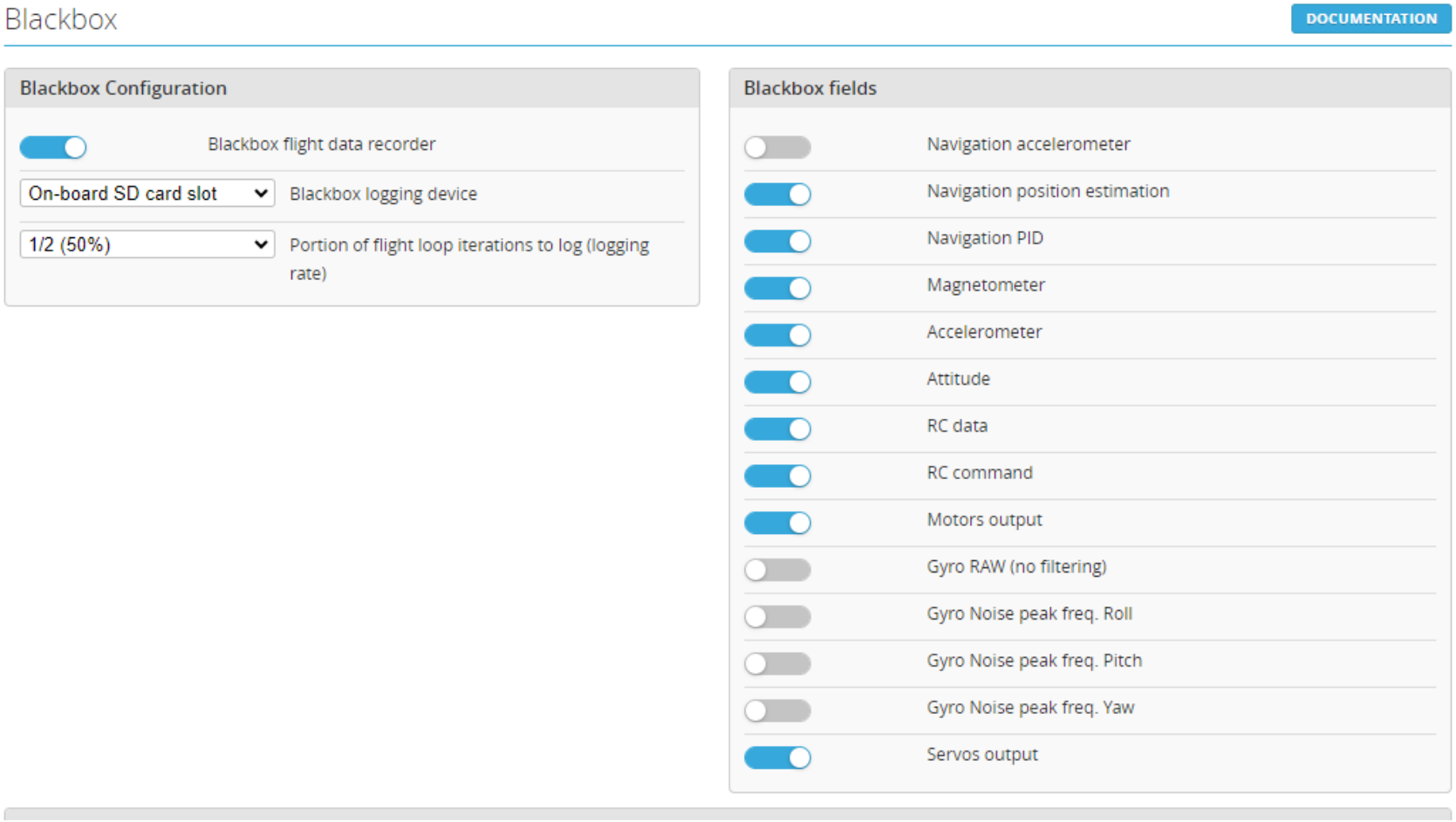
Keep Yaw FF = 0 and turn-assist yaw = off/0 (Mixer/Advanced). This stops the autopilot from constantly "helping" with rudder.

set mag_hardware = NONE
set fw_p_yaw = 0
set fw_i_yaw = 0
set fw_d_yaw = 0
set fw_ff_yaw = 0
save

4. If drift persists with MAG off:

We will Inspect pusher side-thrust. Shim the motor mount (thin washer) to cancel the bias; re-test.

Enable the Blackbox for the next flight



Send me: the .BFL, your diff all, and what you observed.